Task 1

Create a basic singly linked list structure in C. Define a structure for a node that holds an integer value and a pointer to the next node. Implement functions to add nodes at the beginning and end of the list.

Sample Input:

- Add 5 to the beginning.

- Add 10 to the end.

- Add 15 to the end.

Sample Output:

Linked List: 5 -> 10 -> 15

Task 2

Extend your linked list operations. Implement functions to insert a node after a specific value or at a given position and to delete nodes with a given value or at a given position.

Sample Input:

- Insert 25 after 10.

- Delete the node with the value 10.

- Insert 20 at position 2.

- Delete the node at position 3.

Sample Output:

Linked List: 5 -> 25 -> 20

Task 3

Learn to reverse the linked list in-place. Implement a function that changes the order of nodes in the list.

Sample Input:

- Reverse the linked list.

Sample Output:

Linked List: 20 -> 25 -> 5

Task 4

Implement a function to check if a linked list contains a cycle.

Sample Input:

- Create a linked list with a cycle.

Sample Output:

Has Cycle: Yes

Cycle Start Node: 10

Task 5

Build on your knowledge by merging two sorted linked lists into one sorted list. Implement a function to do this while updating the head pointers correctly.

Sample Input:

- Merge two sorted lists: List A: 5 -> 10, List B: 7 -> 12.

Sample Output:

Merged List: 5 -> 7 -> 10 -> 12